

Consumer Goods – Adhoc Business Insights

SQL-Based Business Analysis
Fiscal Year 2020–2021



Business Problem

- **AtliQ Hardwares (imaginary company)** is a leading computer hardware manufacturer operating across multiple countries.
- However, the management lacks quick and actionable insights to make data-driven decisions.
- To strengthen the analytics function, the company conducted a SQL challenge to analyze business performance and generate meaningful insights.

Objective of the Analysis

The objective of this project is to:

- Analyze product growth and portfolio expansion
- Identify key revenue-driving channels
- Understand customer discount strategies
- Evaluate sales performance across months and quarters
- Provide actionable insights to support management decisions

Dataset Overview

The analysis is based on the following datasets:

- **Customer Data (dim_customer)** – Customer details, region, and channel
- **Product Data (dim_product)** – Product categories, segments, and divisions
- **Monthly Sales Data (fact_sales_monthly)** – Sales quantity and fiscal year
- **Gross Price Data (fact_gross_price)** – Product pricing details
- **Manufacturing Cost Data** – Production cost per product
- **Pre-Invoice Discount Data** – Discount percentage offered to customers

Dataset Period: Fiscal Year 2020 & 2021

Adhoc Request 1 – Markets of Atliq Exclusive (APAC).

Business Question

In which APAC markets does customer “Atliq Exclusive” operate?

Output Summary.

market
India
Indonesia
Japan
Philippines
South Korea
Australia
Newzealand
Bangladesh

Key Insight:

Atliq Exclusive operates across **8 APAC markets**, showing a strong regional presence.

Adhoc Request 2 – Unique Product Growth (2020 vs 2021).

Business Question

What is the percentage increase in unique products in 2021 compared to 2020?

Output Summary

	unique_products_2020	unique_products_2021	percentage_chg
▶	245	334	36.33

Key Insight:

There is a noticeable growth in product portfolio in **2021**, indicating **expansion strategy and increased market demand**.

Adhoc Request 3 – Product Count by Segment

Business Question

What is the unique product count for each segment?

Output Summary

segment	product_count
Notebook	129
Accessories	116
Peripherals	84
Desktop	32
Storage	27
Networking	9

Key Insight:

Notebook segment has the highest product count, indicating that the company is heavily focused on expanding its notebook product line compared to other segments.

Adhoc Request 4 – Segment-wise Product Increase (2021 vs 2020).

Business Question

Which segment had the highest increase in unique products in 2021 compared to 2020?

Output Summary.

	segment	product_count_2020	product_count_2021	difference
	Accessories	69	103	34
	Notebook	92	108	16
	Peripherals	59	75	16
	Desktop	7	22	15
	Storage	12	17	5
	Networking	6	9	3

Key Insight:

Accessories segment recorded the highest increase in unique products (+34) in 2021 compared to 2020, indicating a strong expansion focus in this category and higher strategic investment in accessory products.

Adhoc Request 5 – Highest & Lowest Manufacturing Cost

Business Question

Which products have the highest and lowest manufacturing costs?

Output Summary

	product_code	product	manufacturing_cost
▶	A2118150101	AQ Master wired x1 Ms	0.8920
	A6120110206	AQ HOME Allin1 Gen 2	240.5364

Key Insight:

AQ HOME Allin1 Gen 2 has the **highest manufacturing cost (240.54)**, while **AQ Master wired x1 Ms** has the **lowest (0.89)**, showing a significant cost variation across product categories. This indicates that premium hardware products require substantially higher production investment compared to basic accessory products.

Adhoc Request 6 – Top 5 Customers by Discount (India, 2021).

Business Question

Which top 5 customers received the highest average pre-invoice discount in FY 2021 (India)?

Output Summary

	customer_code	customer	average_discount_percentage
►	90002009	Flipkart	0.31
	90002006	Viveks	0.30
	90002002	Croma	0.30
	90002003	Ezone	0.30
	90002016	Amazon	0.29

Key Insight:

Flipkart received the highest average discount (31%) in FY 2021, followed closely by **Viveks**, **Croma**, and **Ezone (30%)**. This indicates that the company offers higher discounts to large retail and e-commerce customers to strengthen partnerships and drive higher sales volumes.

Adhoc Request 7 – Monthly Gross Sales (Atliq Exclusive).

Business Question

What is the monthly gross sales amount for Atliq Exclusive?

Output Summary

Month	Year	Gross_sales_amount
9	2019	4496259.67
10	2019	5135902.35
11	2019	7522892.56
12	2019	4830404.73
1	2020	4740600.16
2	2020	3996227.77

Key Insight:

Gross sales peaked in **November 2019 (7.52M)**, indicating strong festive season performance. However, a sharp decline is observed from February 2020 onwards, suggesting a major business disruption affecting sales performance.

Adhoc Request 8 – Quarterly Sold Quantity (2020).

Business Question

Which quarter had the maximum total sold quantity in 2020?

Output Summary

Quarter	total_sold_quantity
Q1	7005619
Q2	6649642
Q4	5042541
Q3	2075087

Key Insight:

Q1 recorded the **highest total sold quantity (7.0M)**, indicating peak demand during the early fiscal months. In contrast, **Q3** showed the **lowest performance**, suggesting seasonal slowdown in sales.

Adhoc Request 9 – Channel Contribution (FY 2021).

Business Question

Which sales channel contributed the highest gross sales in FY 2021?

Output Summary

	channel	gross_sales_mln	percentage
▶	Retailer	1219.08	73.23
	Direct	257.53	15.47
	Distributor	188.03	11.30

Key Insight:

The **Retailer channel** contributed the **highest gross sales (1219.08M)**, accounting for **73.23%** of total revenue in FY 2021. This indicates a strong dependency on the retailer network for revenue generation and market reach.

Adhoc Request 10 – Top 3 Products by Division (FY 2021).

Business Question

What are the top 3 products in each division based on total sold quantity in FY 2021?

Output Summary.

division	product_code	product	total_sold_quantity	rank_order
N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
N & S	A6818160202	AQ Pen Drive DRC	688003	2
N & S	A6819160203	AQ Pen Drive DRC	676245	3
P & A	A2319150302	AQ Gamers Ms	428498	1
P & A	A2520150501	AQ Maxima Ms	419865	2
P & A	A2520150504	AQ Maxima Ms	419471	3
PC	A4218110202	AQ Digit	17434	1
PC	A4319110306	AQ Velocity	17280	2
PC	A4218110208	AQ Digit	17275	3

Key Insight:

N & S (Pen Drives) and **P & A (Mouse)** products dominate sales in FY 2021, while the **PC division** shows significantly lower sales volumes.

Final Business Summary.

- Strong APAC presence across 8 markets
- Retail channel drives majority revenue (73%)
- Accessories significantly outperform the PC division
- Q1 recorded the highest sales performance
- Strategic discounts used to retain key customers